Dr. Soly Mathew

Materials Scientist | Flexible Sensor Researcher | ORCID: 0009-0001-2480-4682
Email: soly.mathew@vit.ac.in | Portfolio: soly-mathew.github.io | Dept. of Physics, VIT, India

ACADEMIC & RESEARCH APPOINTMENTS

Scientific Officer - Department of Art Conservation, Indian Institute of Heritage, Delhi
Jul 2026 - Present | Materials science analysis, degradation characterisation, and conservation research.

Project Associate - Indian Institute of Technology Jodhpur (IIT Jodhpur)
Oct 2025 - Jun 2026 (9 mos) | Advanced functional nanocomposites, sensor design, and electromechanical evaluation.

Assistant Professor (Adhoc) - NSS College of Engineering, Palakkad
Aug 2025 - Oct 2025 (3 mos) | Engineering physics lectures, laboratory supervision, and undergraduate mentoring.

EDUCATION

Doctor of Philosophy (PhD) in Materials Science (2020 - Present)
Vellore Institute of Technology (VIT), School of Advanced Sciences, Department of Physics, India
Supervisor: Dr. Krishnamoorthi Chintagumpala | Focus: Polymer nanocomposites & flexible capacitive pressure sensors

VERIFIED PEER-REVIEWED JOURNAL PUBLICATIONS (5)

1. Smart Materials and Structures (2026) - DOI: 10.1088/1361-665X/ae8228
   PDMS-TiO2-ZnO ternary dielectric composite based flexible capacitance pressure sensors (IOP Publishing)
2. Journal of Applied Polymer Science (2026) - DOI: 10.1002/app.70439
   Highly Sensitive Capacitance Pressure Sensor With ZnO-GO-PVDF Dielectric Composite Layer (Wiley)
3. Flexible and Printed Electronics (2025) - DOI: 10.1088/2058-8585/adc81e
   Design and characterization of PDMS-CB-ZnO ternary composite flexible pressure sensors (IOP Publishing)
4. Sensors International (2025) - DOI: 10.1016/j.sintl.2025.100330
   Zirconia nanoparticles filled PDMS composites based flexible capacitance pressure sensor (Elsevier)
5. IEEE Access (2025) - DOI: 10.1109/ACCESS.2025.3577066
   Fabrication and Evaluation of PDMS-La2O3 Composites Flexible Capacitance Pressure Sensors (IEEE)

CORE EXPERIMENTAL, ML & COMPUTATIONAL COMPETENCIES

Material Synthesis: Solution blending, probe sonication, shear homogenization, surface silanization, vacuum degassing.
Device Fabrication: Dynamic spin-coating, automated doctor-blade casting, cleanroom thermal polymerization, lamination.
Characterisation: FESEM microscopy, EDX elemental mapping, X-ray Diffraction (XRD), FTIR molecular spectroscopy.
Electromechanics: Broadband Dielectric Spectroscopy (BDS, 100 Hz-1 MHz), high-frequency LCR, cyclic compression (>2000 cycles).
Materials Science ML: Scikit-learn (Random Forest, SVR, Gradient Boosting), feature engineering, sensor sensitivity prediction.
Scientific Computing: Python 3 (NumPy, Pandas, SciPy, Matplotlib), JupyterLab, Claude Code, Copilot, Antigravity, pytest, Git.

RESEARCH REFEREES & COLLABORATION INQUIRIES

Doctoral Referee: Dr. Krishnamoorthi Chintagumpala (Associate Professor, Dept. of Physics, VIT University)
Target Institutional Alignment: Danish Technological Institute (DTI), FORCE Technology, Nordic MedTech & European R&D.